

of infrared spectrum and visible light to track movement at a high speed. Thereafter, the software written on top of the hardware performs background subtraction; elimination and detection that helps the system differentiate between the projection in motion and the real person movement. This makes the product highly accurate giving a good user experience.

"Our hardware, with MagixSense sensors as the main ingredient, coupled with software, has helped us achieve high precisions. The sensor technology provides high-quality multi-point gesture recognition. Customers can link up to four projectors and four sensors on one computer, creating larger projection areas with a single licence and application," says Tapadia.

The MagixTouch product too has elevated the multi-touch user experience to another level. "It creates multitouch

intelligence has been embedded that helps the sensor distinguish between the hand, feet, body and head. Our hardware detects better images at a fast speed. Based on that, we use different classification algorithms that allow us to categorise different parts of the body and build skeleton structures for mapping and tracking body parts. This helps the sensor recognise the hand, body, head and feet in separate frames and provide the information to the developers, who, in turn, can develop applications around the movement of different body parts too," explains Tapadia.

Suitable for any commercial environment. Installing and running such interactive systems could face challenges like changing light conditions and irregular surfaces in a commercial environment. Has TouchMagix been able to overcome them?

"Yes, we have. Taking this project

includes a creative platform with loads of pre-designed templates along with source code available for customisation, a downloadable open software development kit (SDK) for quick start of content development and their own customisation services.

"The easy-to-use creative platform which requires just graphics and content knowledge is available free of cost. The open SDK allows developers to get coordinate feeds, etc, helping them develop applications from scratch. Lastly, we offer customisation services that cost between Rs 10,000 and Rs 100,000 depending on the complexity of customisation," says Tapadia.

TouchMagix also claims to be cost-effective compared to competing products. "Its cost ranges from Rs 300,000 for a basic system to Rs 600,000 for a full-fledged system. Competing products cost way above that," says Tapadia.

The perfect advertising medium.

TouchMagix comes with features like a remote scheduler and Web analytics that make it a perfect advertising medium. "The remote scheduler allows one to schedule content remotely and keep track of it. So as a brand you get the freedom to decide the exact time and the number of times the ad should appear. Further, it has analytics like people counting tool that provides data on the number of interactions with the surface and the average time spent by each person interacting with the surface," says Tapadia.

Roadmap ahead!

Looking ahead, Tapadia and his team hope to add advanced motion and touch modules to their existing system. "We recently launched the clapsensing module built for consumer engagement games. We shall soon be adding mobile interfaces wherein one can connect to the interactive system via the mobile. We shall keep adding motion and touch modules," informs Tapadia. ●

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TouchMagix doesn't detect the full body as a single interactive zone. Intelligence has been embedded that helps the sensor distinguish between the hand, feet, body and head

interaction displays ranging from 81.3 cm (32 inches) to 3.8 metres (150 inches). Instead of using the same camera chips for touch sensing, we chose optical touch technology that could first turn any surface, be it an LCD or projected surface, into a multi-touch system by just mounting a glass on the surface. Further, the multi-touch technology allowed us to do 40-point simultaneous touch detection on a surface. These touches are tracked at 50 frames per second rate. This provides a brilliant user experience," says Tapadia.

Ability to distinguish between parts of the human body. Further, TouchMagix has intelligence embedded into it that allows it to distinguish between movements by different parts of the body.

"Unlike competing products, TouchMagix doesn't detect the full body as a single interactive zone. Intel-

from our labs to the commercial grid environment involved challenges regarding robustness, lighting, etc. Our automated algorithms for background sensing, rebuilding of backgrounds and adapting to the changing light conditions provide the robustness needed for a commercial environment. Further, our sensor elimination system tackles the ambient light problem and automatically changes according to the ambient light. The three installations in the Mumbai airport that are used as 24/7 advertising media by multiple agencies speak of our success in overcoming such issues," says Tapadia. They also couple with third-party software vendors for geometric corrections to achieve seamless user experience even on irregular surfaces.

Large content suite and cost-effective. Further, TouchMagix offers a rich content development package, which